BT301 SUBJECTIVES

MID TERM- 100%

Introduction-to-Biotechnology

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SUBJECTIVES

Define primary, secondary, tertiary structure of protein:

Primary structure involves the linear sequence of amino acids in a polypeptide chain. Secondary structure results from interactions such as alpha-helices and beta-sheets, folding due to hydrogen bonding. Tertiary structure represents the overall three-dimensional arrangement of a single polypeptide chain. Quaternary structure involves the interaction of multiple polypeptide chains in a protein complex. These hierarchical levels collectively determine a protein's function and are crucial for its stability and biological activity.

Function of chloroplast genes:

Chloroplast genes serve vital functions, categorized into two groups: those facilitating replication, transcription, and translation processes, and those integral to photosynthesis. These genes collectively orchestrate the chloroplast's ability to sustain cellular energy production and metabolic activities within plant cells.

Role of electroporation technique:

Electroporation is a rapid and straightforward method for introducing cloned genes into diverse microbial, plant, and animal cells, such as E. coli. This technique involves applying an electric field to cells, temporarily increasing the permeability of the cell membrane. This heightened permeability allows the uptake of DNA, facilitating efficient gene transfer and making electroporation a valuable tool in genetic engineering and molecular biology applications across various organisms.

Chemical composition of DNA:

DNA, or deoxyribonucleic acid, is composed of nucleotide units. Each nucleotide consists of a phosphate group, a deoxyribose sugar molecule, and one of four nitrogenous bases: adenine (A), thymine (T), cytosine (C), or guanine (G). The sequence of these bases forms the genetic code. Adenine pairs with thymine, and cytosine pairs with guanine through hydrogen bonds, creating the double-stranded helical structure of DNA. This unique composition allows DNA to store and transmit genetic information, playing a fundamental role in heredity and cellular processes.

Principles of western blotting:

Western blotting involves separating proteins by electrophoresis, transferring them to a membrane, and probing with specific antibodies to detect target proteins. This technique enables the identification and quantification of proteins based on their size and antibody binding.

Rm system long:

Restriction-Modification is a bacterial defense mechanism against foreign DNA. Restriction enzymes recognize specific DNA sequences and cleave them, while modification enzymes

methylate the host DNA at the same recognition site, protecting it from the restriction enzyme's activity.

Types: The Restriction-Modification (R-M) system encompasses four types: Type I: Type I systems were the first to be characterized from E. coli K 12. The active enzyme consists of two restriction subunit, two modification subunit and one recognition subunit. Type II: Most of the useful R- M system is Type II. Type II enzymes recognize a defined sequence and cut within it. Type III: Type III enzymes have symmetrical recognition sequences but otherwise resemble type I systems and are of little value. Type Iis: Type IIs systems have similar cofactors and structure to type II but restriction occurs at a distance from recognition site that limits their usefulness.

Role of ori in region of plasmid:

The Ori (origin of replication) region in a plasmid is crucial as it serves as the starting point for DNA replication. This ensures the plasmid's duplication during cell division, allowing its propagation and maintenance in host cells. The precise initiation of replication from the Ori ensures the faithful transmission of genetic material. Understanding and manipulating the Ori is fundamental in molecular biology, enabling controlled plasmid replication for various applications, such as gene cloning, recombinant DNA technology, and biotechnological research.

Define the major and minor grooves and their length:

In DNA, the minor groove is a narrow space between double helical strands, and its length is about 3.4A0. The major groove, is a wide space between double helical strands, and its length about 36A0. Both grooves play roles in genetic regulation and protein interactions.

How large DNA are packed be in small volume:

To efficiently pack the considerable length of DNA within a eukaryotic cell's small volume, DNA molecules are intricately coiled and tightly wound around histone proteins, forming chromatin. These nucleosome structures consist of approximately 200 base pairs of DNA and nine histones, creating a compact and organized unit. During cell division, chromatin condenses further into mitotic chromosomes, reducing the DNA length by about 50,000 times. The condensed heterochromatin state contrasts with the more extended euchromatin, showcasing dynamic regulation of genetic material packaging in eukaryotes.

Sequencing genomes in genetic variation and its implications on human health:

Sequencing genomes is crucial for understanding genetic variation, gene numbers, locations, and functions. It reveals DNA sequence, genome organization, and noncoding DNA types, aiding in studying gene regulation, protein interactions, and evolutionary conservation. Genome sequencing correlates single nucleotide polymorphisms (SNPs) with health, predicting disease susceptibility and identifying genes related to complex traits. It advances diagnostics through systems biology and enables therapeutic discoveries such as drug targets, rational drug design, and gene therapies, fostering a comprehensive understanding of metabolism and disease mechanisms for improved human health.

Genetic variation:

Genetic variation refers to the diversity of genetic material within a population or species. It arises from differences in DNA sequences, including mutations, recombination, and other genetic processes, influencing individual traits and contributing to the evolution and adaptation of organisms.

Advantage of low molecular weight vector:

Low molecular weight vectors offer advantages in gene delivery, as they can more efficiently penetrate cell membranes. Their smaller size facilitates easier manipulation, packaging, and transportation, enhancing the effectiveness of gene transfer and the development of various biotechnological applications.

Tertiary structure of protein:

The overall three dimensional arrangement of all atoms in a protein is referred to as the protein’s tertiary structure. It includes longer range aspects of amino acid sequence. Amino acids that are far apart in the polypeptide chain may interact within the completely folded structure of a protein. Interacting segments of polypeptide chains are held in their characteristic tertiary

positions by different kinds of weak interactions (and sometimes by covalent bonds) between the segments. Large polypeptide chains usually fold into two or more globular clusters known as domains, which often give these proteins a bi-or multilobal appearance.

Types of RNA:

RNA, or ribonucleic acid, comes in various types with distinct roles. Messenger RNA (mRNA) carries genetic information from DNA to ribosomes for protein synthesis. Transfer RNA (tRNA) transports amino acids to ribosomes during protein assembly. Ribosomal RNA (rRNA) is a structural component of ribosomes. Small nuclear RNA (snRNA) aids in RNA splicing, and small nucleolar RNA (snoRNA) guides chemical modifications in ribosomal RNA.

X ray diffraction analysis of DNA:

X-ray diffraction analysis of DNA involves exposing a crystalized DNA sample to X-rays, resulting in diffraction patterns. The scattered X-rays produce a distinctive pattern revealing the spatial arrangement of atoms in the DNA molecule. This technique, pioneered by Rosalind Franklin and Maurice Wilkins, played a pivotal role in elucidating the double helical structure of DNA, as famously discovered by Watson and Crick.

Green Biotechnology with example:

Green Biotechnology involves the use of plant cells, tissues, or whole plants to develop beneficial products or solve environmental issues. An example is genetically modified crops, such as insect-resistant Bt cotton, which enhances agricultural sustainability and reduces the need for chemical pesticides.

Processes of comparative genomics:

Comparative genomics analyzes genetic information across different species to identify similarities, differences, and evolutionary relationships. It involves genome sequencing, annotation, and the comparison of gene content, structure, and function, providing insights into evolutionary processes and genetic diversity.

Hershey-Chase Experiment for discovery of DNA:

The Hershey-Chase experiment (1952) confirmed that DNA, not protein, carries genetic information. They used T2 bacteriophage to infect Escherichia coli, demonstrating that only DNA, labeled with radioactive phosphorus, entered bacterial cells and directed viral replication, providing crucial evidence for the role of DNA in heredity.

Benefits of genome research in molecular medicine:

Genome research in molecular medicine brings significant benefits by enhancing disease diagnosis, detecting genetic predispositions (e.g., cancer, diabetes), and enabling the creation of molecularly informed drugs. It also supports advancements in gene therapy and control systems, offering innovative approaches to disease treatment and contributing to the development of personalized and precision medicine strategies for improved healthcare outcomes.

Difference between pBR322 and pBR325:

pBR322 and pBR325 are plasmids used in genetic engineering. pBR322 is an in vitro constructed cloning vehicle with ApR and TcR genes from RSF2124 and pSC101, and a replication element from pMB1. It has 40 restriction enzyme target sites but is less suitable for long DNA segments. In contrast, pBR325, a derivative of pBR322, also encodes chloramphenicol resistance along with ampicillin and tetracycline, offering expanded applications and improved vectors like pUC19 and pAT153. These plasmids have been crucial in DNA cloning and genetic engineering experiments, contributing to advancements in molecular biology research.

Primary structure of protein note:

The primary structure of a protein is its linear sequence of amino acids linked by peptide bonds. This sequence is dictated by the genetic code. The unique order of amino acids determines the protein's specific structure and function.

Alpha-beta sheet character or role in secondary structure of protein:

The alpha-beta sheet is a common motif in the secondary structure of proteins. It consists of alternating alpha helices and beta strands, forming a stable and compact structure. This arrangement imparts mechanical strength and stability to proteins, crucial for their biological

functions. The beta strands contribute to the sheet's rigidity, while the alpha helices provide flexibility. The alpha-beta sheet is found in diverse proteins and plays a vital role in maintaining structural integrity and facilitating molecular interactions within biological systems.

PCR role function in genome:

Polymerase Chain Reaction (PCR) is a pivotal technique in genomics. It amplifies specific DNA sequences, enabling the rapid and precise replication of target genes. PCR has revolutionized genome research by facilitating DNA sequencing, mutation analysis, and gene cloning. Its applications range from diagnostics to forensics, allowing the detection and study of genetic variations. PCR's efficiency in replicating DNA has significantly advanced our understanding of genomes, gene functions, and various molecular processes in genetics and molecular biology.

Structure of mRNA:

mRNA (messenger RNA) is a single-stranded nucleic acid with a ribose sugar backbone. It may be monocistronic or polycistronic. It carries genetic information from DNA to the ribosome for protein synthesis. The sequence of nucleotide bases, adenine, cytosine, guanine, and uracil, determines the encoded genetic instructions in the form of codons.

Write a note on PCR:

Polymerase Chain Reaction (PCR) is a revolutionary molecular biology technique that amplifies specific DNA segments. It involves repeated cycles of denaturation, primer annealing, and extension using a DNA polymerase enzyme. PCR enables the exponential replication of target DNA, making it a powerful tool for applications such as gene cloning, DNA sequencing, and diagnostics. Its speed, precision, and versatility have transformed genomics, allowing the study of genetic variations, gene expression, and forensic analysis, with broad implications in research, medicine, and biotechnology.

Side effects of ethidium bromide when in contact with body:

Ethidium bromide, a commonly used DNA-staining agent, can pose health risks. Exposure may cause skin and eye irritation, and inhalation can lead to respiratory issues. Ethidium bromide is a potential mutagen and carcinogen, emphasizing the importance of proper handling, protective measures, and disposal procedures in laboratories.

Purpose of northern blotting technique:

Northern blotting is used to study gene expression by detecting and analyzing RNA molecules. It involves electrophoretically separating RNA samples, transferring them to a membrane, and hybridizing with labeled probes to identify specific RNA transcripts, providing insights into gene activity and regulation.

Types of blotting techniques:

Blotting techniques are molecular biology methods for transferring biomolecules from a gel to a membrane for further analysis. Southern blotting detects DNA, Northern blotting identifies RNA, and Western blotting targets proteins. Each type involves electrophoretic separation, transfer to a membrane, and specific detection methods for the respective biomolecules.

DNA is polymer of deoxyribonucleotide. Enlist and explain the chemical composition of deoxyribonucleotide:

A deoxyribonucleotide, the monomer of DNA, consists of three components: a deoxyribose sugar molecule, a phosphate group, and one of four nitrogenous bases (adenine, thymine, cytosine, or guanine). The deoxyribose sugar forms the backbone, linked by phosphodiester bonds between adjacent nucleotides. The nitrogenous base, either a purine (adenine or guanine) or pyrimidine (thymine or cytosine), connects to the sugar. This composition imparts the unique sequence and structure to DNA, encoding genetic information and facilitating various cellular processes.

How comparative genomics find conserved gene in different species:

Comparative genomics identifies conserved genes across species by analyzing DNA sequences from different organisms. Computational algorithms align genomic data, revealing regions with high sequence similarity, suggesting functional importance. Conserved genes are evolutionarily retained due to their crucial roles, indicating shared biological functions. Comparative analysis identifies conserved motifs, regulatory elements, and coding sequences, aiding in understanding

gene function, evolution, and potential therapeutic targets. The conservation of genes provides insights into fundamental biological processes and highlights key elements shaping the diversity of life.

Which pregnancy test was used to test the gender in ancient period?

A papyrus described a test in which a woman who might be pregnant could urinate on wheat and barley seeds over the course of several days: “If the barley grows, it means a male child. If the wheat grows, it means a female child. If both do not grow, she will not bear at all.

Which factors affect the mobility of DNA in gel ?

DNA mobility in gel electrophoresis is influenced by factors such as DNA size, gel concentration, and voltage applied. Smaller DNA fragments migrate faster, higher gel concentrations provide greater resolution, and increased voltage expedites migration, affecting the separation and visualization of DNA bands.

Components of Bacterial genome:

Bacterial genomes are characterized by high coding density, with 85-90% of DNA encoding genes. Prokaryotes exhibit a gene frequency of 1 gene per 1000 bases. A bacterial genome consists of a single circular DNA molecule containing essential components. The chromosome carries genes for vital cellular functions, and plasmids, additional circular DNA, may harbor accessory genes. Regulatory elements control gene expression, ensuring proper cellular processes. Bacterial genomes lack introns, and operons organize related genes. Mobile genetic elements like transposons contribute to genetic diversity. Understanding these components is crucial for studying bacterial genetics, evolution, and developing therapeutic interventions.

What are protomer, oligomer and multimer?

A protomer is a single unit of a protein or macromolecule. Oligomer refers to a complex composed of a few protomers, while a multimer consists of numerous protomers. These terms describe the organization of protein subunits, reflecting the structural and functional complexity of macromolecular assemblies in biological systems.

What is difference between circular DNA and Open circle or OC DNA?

Circular DNA is a complete, covalently closed loop, while open circle (OC) DNA is a partially cleaved circular DNA with a break or nick in one strand. OC DNA results from enzymatic or chemical treatments and is distinct from the continuous, closed structure of circular DNA.

What is biotechnology and waited names of it's various branches?

Biotechnology is the use of microbes, animal/plant cells and their products to synthesize, break down or transform materials. Primarily it includes the use of recombinant DNA technology and genetic engineering techniques to improve upon the quality of processes.

1. Medical Biotechnology
2. Agricultural Biotechnology
3. Industrial Biotechnology
4. Environmental Biotechnology
5. Forensic Biotechnology

Chargaff’s rule:

Chargaff's rule, formulated by biochemist Erwin Chargaff, states that in DNA, the amounts of adenine (A) are equal to thymine (T), and the amounts of guanine (G) are equal to cytosine (C). This base pairing specificity is fundamental to the double-stranded nature of DNA, as adenine pairs with thymine via two hydrogen bonds, and guanine pairs with cytosine via three hydrogen bonds, ensuring structural stability and accurate replication of genetic information.

Two factors results in high yield of Plasmid DNA:

High yield of plasmid DNA is influenced by efficient bacterial culture growth and optimized plasmid extraction methods. Achieving optimal cell density during culture and employing suitable extraction protocols enhance plasmid DNA yield, crucial for various molecular biology applications.

Purpose of Use of Southern blotting:

Southern blotting is used to detect specific DNA sequences in a sample. It involves electrophoretic separation of DNA fragments, transfer to a membrane, hybridization with a labeled probe, and autoradiography, allowing identification and analysis of target DNA sequences, aiding in gene mapping, and genetic studies.

Write Pauling and Corey theory about peptide bond formation in primary protein structure?

Linus Pauling and Robert Corey's analysis of the peptide bond formed the basis for understanding protein structure. They determined that the peptide C-N bond is shorter than in a simple amine. The six atoms of the peptide group are coplanar, and due to partial double-bond character, C-N bonds are rigid and planar, limiting free rotation. Pauling and Corey introduced bond angles (φ and ψ) to describe allowed rotations, laying the groundwork for comprehending primary protein structure and subsequent research in protein science.

Define genomics it's benefits?

Genomics is the study of complete sets of genes and their functions within an organism. Its benefits include insights into genetic variation, disease susceptibility, personalized medicine, agriculture improvement, and advancements in diagnostics and therapeutics, fostering a deeper understanding of biological systems for various applications.

Characteristics of bacterial genome?

Bacterial genomes are typically single, circular DNA molecules with high coding density. Prokaryotic genomes exhibit 1 gene per 1000 bases, and there is significant variation in size between bacterial species. Some bacterial genomes, like Tremblaya princeps, can be relatively small (140kb), while others, like Sorangium cellulosum, are larger (14000kb).

Advantages of genome Sequencing:

Genome sequencing provides insights into genetic variation, disease causes, and drug responses. It aids in understanding evolutionary relationships, functional genomics, and personalized medicine. It enables precise diagnostics, facilitates crop improvement, and contributes to advancements in biotechnology and medical research.

Gauecher disease:

Gaucher disease is a rare genetic disorder where glucocerebroside accumulates in cells due to a deficiency of the enzyme glucocerebrosidase. This buildup leads to organ and tissue damage. Symptoms include enlarged liver and spleen, anemia, and bone abnormalities. Enzyme replacement therapy is a common treatment.

Genome definition and function:

A genome is the complete set of genes in an organism's DNA. It contains instructions for the organism's growth, development, and functioning. Genomes encode genetic information, determining traits and facilitating cellular processes for the maintenance and reproduction of living organisms.

Genome characteristics in Long:

Genomes, comprising DNA or RNA, contain an organism's genetic information. They are typically organized into chromosomes and vary in size, structure, and complexity. Genomes exhibit specific base pair sequences, encode genes for cellular functions, and include non-coding regions. Genetic variation within genomes contributes to biodiversity. Genomes undergo replication, transcription, and translation processes. Factors like coding density, gene frequency, and mobile genetic elements impact genome characteristics, reflecting the diversity of life and playing crucial roles in evolution and inheritance.

How many functional group attach to the carbon atom including R group:

In organic compounds, a carbon atom typically forms four bonds, including its interactions with functional groups and the R group (side chain). Functional groups are specific arrangements of atoms that impart characteristic chemical properties to molecules. These groups, such as hydroxyl (–OH), amino (–NH2), and carbonyl (C=O), are attached to the carbon backbone. The R group, unique for each organic molecule, determines the compound's identity, influencing its chemical behavior and properties.

Chloroplast gene function

Chloroplast genes encode proteins, tRNAs, and rRNAs essential for chloroplast function in photosynthesis. They contribute to the synthesis of pigments, electron transport chain components, and enzymes, playing a crucial role in energy conversion and the production of organic molecules in plant cells.

Plasmid genes function:

Plasmid genes often carry traits like antibiotic resistance or the ability to metabolize specific compounds. They provide a selective advantage to bacteria, aiding survival and dissemination of these traits. Plasmids are used in genetic engineering to introduce desired traits into host organisms.

Watson and crick model of DNA:

The Watson and Crick model for DNA, proposed in 1953, describes the structure of DNA as a double helix. It consists of two anti-parallel strands held together by hydrogen bonds between complementary base pairs (adenine with thymine, and guanine with cytosine). The sugarphosphate backbones run on the outside, while the bases are stacked in the interior. This model provided the foundation for understanding DNA replication, transcription, and genetic information storage.

Define Translation and transcription:

Transcription is the process where RNA is synthesized from a DNA template, forming a complementary RNA strand. Translation is the subsequent process in which the information encoded in the mRNA is used to synthesize a corresponding protein.

Types of clone vector:

Clone vectors, used in genetic engineering, include plasmids, bacterial artificial chromosomes (BACs), yeast artificial chromosomes (YACs), and phage vectors. Plasmids are circular DNA, BACs and YACs carry larger fragments, while phage vectors use bacteriophages to transfer genes.

How useful recombinant DNA technology?

Recombinant DNA technology involves the manipulation and combination of genetic material from different sources. It is invaluable in producing genetically modified organisms, vaccines, and therapeutic proteins, contributing to advancements in medicine, agriculture, and various biotechnological applications.

Steps of cloning?

Cloning with lambda replacement vectors involves isolating and cutting vector DNA, connecting lambda fragments to foreign DNA using DNA ligase, in vitro packaging of recombinant DNA, infecting E. coli cells, isolating phage clones, and checking them for the presence of foreign DNA.

Differ between linker and adaptor?

Linkers and adaptors are both short DNA sequences used in genetic engineering. Linkers connect fragments with compatible ends, while adaptors possess different ends to facilitate the joining of fragments. Both aid in the construction of recombinant DNA molecules.

Work of Franklin and Winlikns?

Rosalind Franklin and Maurice Wilkins contributed to the discovery of the DNA double helix. Franklin's X-ray diffraction images provided critical insights into DNA's structure, and Wilkins collaborated with James Watson and Francis Crick, leading to the elucidation of the DNA model. Their work significantly advanced molecular biology and our understanding of genetics.

Function of DNA:

DNA, or deoxyribonucleic acid, serves as the genetic blueprint for living organisms. Its primary function is to store, transmit, and express genetic information. DNA encodes the instructions for building and maintaining an organism's structure, controlling its growth, development, and functioning.

Four type of cloning strategy:

1. Gene Cloning: Isolating and replicating a specific gene.

2. Reproductive Cloning: Generating an organism with identical genetic material.
3. Therapeutic Cloning: Producing embryonic cells for medical purposes.
4. Molecular Cloning: Creating multiple copies of DNA fragments for analysis or manipulation in molecular biology research.

Difference between prokaryotic and eukaryotic cells:

PROKARYOTIC CELL	EUKARYOTIC CELL
1) Size of the cell is generally small.	1) Size of the cell is large.
2) Centrioles are absent	2) Centrioles are present in animal cells.
3) Nucleolus is absent.	3) Nucleolus is present.
4) Small ribosomes	4) Large ribosomes
5) It contains single chromosome	5) It contains more than one chromosome
6) Mitochondria is absent	6) Mitochondria is present.

How many amino acids are present and give names of 5 amino acids?

There are 20 amino acids that serve as the building blocks of proteins. Five of them are:
1. Alanine (Ala): A nonpolar amino acid.
2. Valine (Val): A branched-chain amino acid.
3. Leucine (Leu): Another essential branched-chain amino acid.
4. Glutamine (Gln): A polar, uncharged amino acid.
5. Lysine (Lys): A positively charged amino acid.

How plants contain genes that act as microbial resistant for antibiotics?

Plants can acquire genes conferring microbial resistance to antibiotics through genetic engineering. Scientists introduce specific antibiotic resistance genes into plant genomes, providing plants with the ability to resist microbial infections and improving their resistance to certain pathogens in agricultural settings.

Explain how two or multiple polypeptide chains arrange in structure of protein

In protein structure, two or multiple polypeptide chains can arrange to form complex quaternary structures. Interactions such as hydrogen bonding, ionic interactions, and hydrophobic forces bring these chains together. The specific arrangement is determined by the protein's primary structure and dictates its function. Examples include hemoglobin, composed of four polypeptide chains, and collagen, formed by three intertwined chains. Quaternary structure enhances the functional diversity and efficiency of proteins in biological processes.

Note on viruses:

Viruses are infectious agents composed of genetic material (DNA or RNA) enclosed in a protein coat. Lacking cellular structure, they rely on host cells to replicate. Viruses infect all forms of life, causing various diseases in humans, animals, and plants. Not considered living entities, viruses bridge the line between living and non-living matter. Understanding viral structures and mechanisms is crucial for combating infections and developing vaccines.

Explain harshy marthachase experiment of discovery of DNA

The Hershey-Chase experiment (1952) confirmed that DNA, not protein, carries genetic information. Martha Chase and Alfred Hershey used T2 bacteriophage labeled with radioactive sulfur (protein) or phosphorus (DNA). Only the phosphorus-labeled DNA entered Escherichia

coli bacteria, leading to viral replication. This demonstrated that DNA, not protein, serves as the hereditary material, solidifying the understanding of DNA's role in genetics and molecular biology.

What happened when super-coiled DNA molecule receives nick:

When a supercoiled DNA molecule receives a nick or single-strand break, its supercoiling is relaxed, converting it into a relaxed or open-circular conformation. This change in topology can affect the stability and functionality of the DNA molecule.

Discribe the structure of DNA on the base of watson and crick:

Watson and Crick proposed the double helix model for DNA in 1953. DNA is composed of two antiparallel strands forming a helical structure. Adenine pairs with thymine, and guanine pairs with cytosine via hydrogen bonds, creating complementary base pairs. The sugar-phosphate backbones run along the exterior. This elegant model explained DNA's ability to store genetic information and undergo replication.

Why Z-DNA is different from the other types of DNA?

Z-DNA differs from other DNA structures due to its left-handed helical orientation, unlike the right-handed helices of B-DNA and A-DNA. Z-DNA forms under specific conditions, and its distinctive zigzag shape arises from alternating purine and pyrimidine base pairs. It has unique structural and functional properties.

Applications of genetic engineering?

Genetic engineering has diverse applications, including the production of genetically modified organisms (GMOs) for agriculture, the development of therapeutic proteins and drugs, gene therapy for treating genetic disorders, and the creation of genetically engineered bacteria for industrial processes. It also plays a crucial role in advancing research in molecular biology, medicine, and biotechnology.